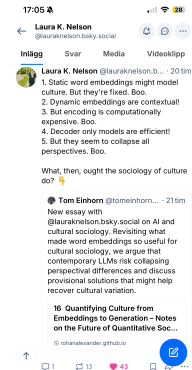


Lecture 4: Large language models (LLMs)/decoder-based models in Social Science

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A pre-lecture plug



LLMs/Decoder-based models in the social sciences

LLMs have become a standard tool for many researchers in everyday life, but there is still no consensus on best practices for social scientific applications.. How *can* and *should* LLMs be used for social science?

People see both opportunities and potential limitation/“dangers” with using LLMs for social scientific research

Opportunities for measurements: text annotation, Törnberg. (2024a),

No “best practices” for how to use LLMs for text annotation, but few-shot and zero-shot learning perform very well (Davidson & Karell, 2025; Törnberg 2024b):

1. Choose an appropriate model
2. Engineer your prompts/Follow a systematic coding procedure/Develop a prompt codebook
3. Validate your model
4. Specify your LLM parameters
5. Discuss ethical and legal implications
6. Examine model stochasticity
7. Consider that your data may be in the training data

Opportunities: **measurement**/text annotation, Törnberg. (2024a), choose an appropriate model

1. Choose an appropriate model

- Reproducibility: use a fixed version of the LLM and ensure that the model will be available in future
- Ethics and legality: e.g., not storing research data compliance with relevant data privacy regulations
- Transparency: documentation of methods, data sources, assumptions, ...
- Culture and language: The LLM should adequately support the language(s) and cultures of your textual data (many English-centric)
- Costs; open-weight (e.g. Llama, Mistral) = free to use, pay-to-use APIs (e.g. OpenAI, Claude).

Opportunities: **measurement**/text annotation, Törnberg. (2024)

2. Engineer your prompts

- Contain the following elements: context (“who answers”), question (what), and constraints (format).
- Give “I don’t know” option
- Ask to indicate (un)certainity

2. Follow a systematic coding procedure

- Iterative process: annotate small sample of data & compare with model output, where do the model get it wrong? Ask LLM to give reason for label. Revise prompt.

2. Develop a prompt codebook (Stuhler et al. 2025)

Opportunities: **measurement**/text annotation, Törnberg. (2024), example prompt

As an expert annotator with a focus on social media content analysis, your role involves scrutinizing Twitter messages related to the US 2020 election. Your expertise is crucial in identifying misinformation that can sway public opinion or distort public discourse.

Does the message contain misinformation regarding the US 2020 election?

Provide your response in JSON format, as follows:

```
{ "contains_misinformation": "Yes/No/Uncertain", "justification": "Provide a brief justification for your choice." }
```

Options:

- Yes
- No
- Uncertain

Remember to prioritize accuracy and clarity in your analysis, using the provided context and your expertise to guide your evaluation. If you are uncertain about the classification, choose 'Uncertain' and provide a rationale for this uncertainty.

Twitter message: [MESSAGE]

Answer:

Opportunities: **measurement**/text annotation, Törnberg. (2024)

3. Validate your model

- Validate after you developed your *final* prompt
- Go beyond accuracy: e.g., F1-score, Cohen's κ , MSE, errors in different subset

4. Specify your LLM parameters: max length (of text), temperature (high $\rightarrow$ more randomness, for classification set low/0), top-p (range of considered token, choose low), etc.

Opportunities: **measurement**/text annotation, Törnberg. (2024)

5. Discuss ethical and legal implications

- Input data can be used for future training (don't feed sensitive or copyright-protected texts)

6. Examine model stochasticity

- Does the same prompt return the same result if run several times? Do small variations in the prompt result in very different results?

7. Consider that your data may be in the training data

- Avoid using publicly available databases (e.g. Manifesto Data) as validation data, they can be in training data

The prompt impacts the output: the “butttterfly” effect, Ye et al. (2026)

Two designs that look interchangeable to a researcher

PLAIN · Prose

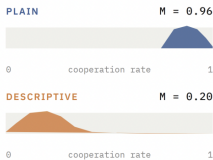
You are a strategic participant focused on maximizing your payoff. Study opponent behavior for exploitable patterns, and adjust your strategy to gain advantage...

DESCRIPTIVE · Bullets

- Maximize your long-term payoff
- Study opponent for exploitable patterns
- Adjust strategy to gain advantage ...

In current practice these would be reported interchangeably, or not reported at all.

...produce opposite macro outcomes



76 pp
COOPERATION RATE GAP

two-agent · gpt-5.2 · $p < 0.001$

...and would support opposite scientific claims

“LLM agents cooperate by default in repeated social dilemmas.”

supportable from PLAIN runs

“LLM agents are strategic exploiters in repeated social dilemmas.”

supportable from DESCRIPTIVE runs

Both findings come from the same model, payoffs, and agents; only the persona format differs.



Opportunities: **silicon samples**, Argyle et al. (2023)

Main idea: Use LLMs to generate survey answers for “respondents” with different social-demographics, works if high **algorithmic fidelity** (AF)

AF = the degree to which patterns of relationships between ideas, attitudes, and sociocultural contexts accurately mirror those within human **subpopulations**

Opportunities: **silicon samples**, Argyle et al. (2023)

Task 1: Use real survey respondents to create a “backstory,” then ask to generate words/text describing partisans. Compare with humans in the survey.

	Describing Democrats	Describing Republicans
Strong Republicans	Ideologically, I describe myself as conservative. Politically, I am a strong Republican. Racially, I am white. I am male. Financially, I am upper-class. In terms of my age, I am young. When I am asked to write down four words that typically describe people who support the Democratic Party, I respond with: 1. Liberal 2. Socialist 3. Communist 4. Atheist.	Ideologically, I describe myself as conservative. Politically, I am a strong Republican. Racially, I am white. I am male. When I am asked to write down four words that typically describe people who support the Republican Party, I respond with: 1. Conservative 2. Male 3. White (or Caucasian) 4. Christian.
Strong Democrats	Ideologically, I describe myself as liberal. Politically, I am a strong Democrat. Racially, I am white. I am female. Financially, I am poor. In terms of my age, I am old. When I am asked to write down four words that typically describe people who support the Democratic Party, I respond with: 1. Liberal. 2. Young. 3. Female. 4. Poor.	Ideologically, I describe myself as extremely liberal. Politically, I am a strong Democrat. Racially, I am hispanic. I am male. Financially, I am upper-class. In terms of my age, I am middle-aged. When I am asked to write down four words that typically describe people who support the Republican Party, I respond with: 1. Ignorant 2. Racist 3. Misogynist 4. Homophobic.

Finding: GPT3 and humans are (almost) indistinguishable

Opportunities: **silicon samples**, Argyle et al. (2023)

Task 3: predict answers to survey given other survey responses relating to socio-demographic variables, attitudes, and behaviors

Finding: GPT-3 captures human-like patterns between different survey items

Conclusion: AF of GPT3 is high enough to use for social science

Opportunities: **silicon samples**, Argyle et al. (2023)

So what?

- Help to design for example survey questions and experimental treatments to guide human research (low cost)
- Give insights into which variables researchers should include in their studies of public opinion
- Generate synthetic data for analyses when there is no real data

But LLMs are different from humans in important ways: variance is smaller than among human resp. (e.g. Bisbee et al. 2024; Xie et al. 2026), some subpopulations are not well represented (Boelaert et al. 2025), less homophilic (Proma et al. 2026), compresses information differently (Shani et al. 2025)

Opportunities: **inference**/simulation studies

Agent-based modeling is a simulation framework used to map micro-level assumptions to macro-level outcomes, but how an agent make decision, who it talks to, what it remembers, and how its preferences change have been hard coded by researchers and are often simple rules

Large language models help us study **more realistic emergent** behaviors of individuals and groups (but also make them more black-box)

Opportunities: inference/simulation studies, e.g. Park et al. (2023)

Give 25 agents:

- A **personality** (prompt with a description of a person; personality traits, jobs, relations), **memory** (store list of personality, relations, previous experience), **reflection** and **planning** ability
- Each iteration they output a text that describes what they do & interact with other agents in local area with a prob.

Emergent social behaviors

- Information diffusion (gossip)
- Relationship memory (ask questions about convos. in past)
- Coordination (one throws party, invite others, that show up)

Opportunities: inference/simulation studies, e.g. Törnberg et al. (2023)

Aim: Test how the level of toxicity in online conversations depend on platform news feed algorithm

- Generate 500 agents with personalities based on survey respondents in the US; the agents “read” newspaper articles and are asked to write a social media comment and/or like the article
- Compare 3 news feed algorithms: (1) interact with posts most liked by friends (30 homophilic ties) (2) interact with post most liked by all users, and (3) interact with posts most liked by people with opposing political view

Opportunities: inference/simulation studies, e.g. Törnberg et al. (2023)

	Toxicity	E-I interpartisan comments	E-I interpartisan likes
Platform 1	0.09	-0.89	-0.97
Platform 2	0.13	-0.70	-0.78
Platform 3	0.07	0.33	-0.18

Table 2. The resulting toxicity and interpartisan interaction.

Results: Platform algorithm impacts toxicity and cross-partisanship interaction

Limitations and potential dangers:

Biases:

- More liberal than conservative, more young than old, more male than female, more extrovert than introvert etc.
- Higher recall than precision (more false positives than false negatives)
- Some LLMs have filters to avoid generating certain content (this changes over time) → not true representations of culture (?)

Errors:

- Hallucinations/make up content, e.g. literature reviews with non-existing papers
- Not better than encoder-only models in many classification tasks

Limitations and potential dangers:

Ethnics:

- Informed consent when exposing real humans to LLMs or LLM-created content
- Working conditions of people working with training LLMs (sweatshops)
- Environmental impact for training LLMs

Replicability:

- Many LLMs updated continuously and will produce new answers over time

Summary

Using decoders for text annotation show great potential, but encoders are likely better for many social scientific concepts (Ollion et al. 2023, Sheni et al. 2026)

Decoders may be used to perform better/less costly surveys, but with many limitations!

Can be used in simulation/experimental settings to simulate human values, conversations, and behaviors

Many more uses are likely to emerge, but still unclear when and how to use LLMs...TBD

Thank you!

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